

What is a Program?

- A program is a sequence of instructions/commands/stmts.
- What are the different kinds of instr^{ns}/stmts in a C program?

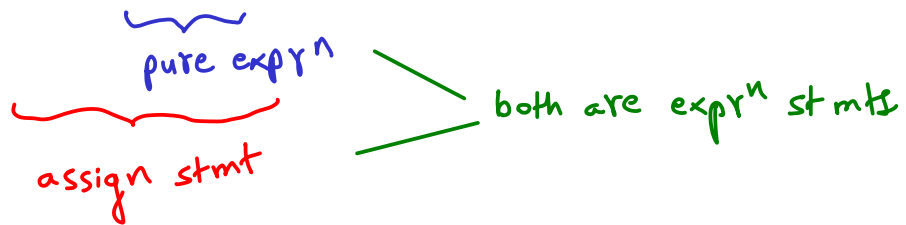
Types of Statements in C:

- Null/empty stmts (just a semicolon (;) indicates a null stmt)
- Exprⁿ stmts
 - Input stmts (e.g: scanf, getchar)
 - Output stmts (e.g: printf, putchar)
 - Pure expr^s (e.g: 3+5; x&& y || 2)
 - Assign stmts (e.g: x=5; a=4+5-y)
- Declaration stmts
 - Variable declⁿ stmts (eg: int a; int b=12)
 - Function declⁿ stmts (eg: int sum(int a, int b))
- If-else stmts / conditional stmts / selection stmts (e.g: if-else; switch)
- Iterative stmts / Looping stmts (eg: for, while, do-while)
- Jump stmts (e.g: break; continue; return; exit)

- Compound stmts
 - ```
{
 s1;
 s2;
 ⋮
 sn;
}
```
  - every individual stmt in a compound stmt should end with a semicolon
  - no need to put a semicolon after the closing bracket

- An  $\text{expr}^n$  connects one or more constants or variables with one or more operators.
- An  $\text{expr}^n$  can be part of a stmt, or it can be a stmt itself.

e.g1:  $x = 2 + 3;$



Note: Among all kinds of stmts, only  $\text{expr}^n$  stmts have/return a value.

- For instance, in e.g1, the value of the pure  $\text{expr}^n$   $2 + 3$  is 5

Similarly the value of the entire assign stmt,  $x = 2 + 3$  is also 5.

So, not only the variable  $x$ , but the entire assign stmt has the value 5.

It is for this reason that this entire stmt can be part of another  $\text{expr}^n$  stmt.

eg2: `printf("%d", x = 2 + 3);` // 5 will be printed

eg3: `y = x = 2 + 3;` // Here  $y$ 's value will be 5,

as well as the entire stmt's value will also be 5.

- `printf` and `scanf` are also  $\text{expr}^n$  stmts, hence they return some values.
- `printf` returns the no. of characters printed while `scanf` returns the no. of values successfully scanned from the input.

eg4: `int x = printf("hello");` //  $x$  value will be 5

eg5: `int x = printf("hello\n");` //  $x$  value will be 6

eg6: `int x = scanf("%d%f", &p, &q);` //  $x$  value will be 2 if both  $p$  and  $q$  are successfully scanned from the i/p

Some points to note while using scanf:

- There are 3 kinds of whitespace characters: space, tab, newline
- In a scanf stmt, if the delimiter is a whitespace character or if no delimiter is specified between two format specifiers, the inputs can be separated by any no. of whitespace characters.

eg1: `scanf ("%d %d", &x, &y);`

eg2: `scanf ("%d %d", &x, &y);`

eg3: `scanf ("%d\t %d", &x, &y);`

eg4: `scanf ("%d\n %d", &x, &y);`

/\* In all these cases, the inputs can be separated by any no. of whitespace characters of any kind \*/

- But if you want the inputs to be separated by a non whitespace character, Say a comma or semicolon, then you need to specify them explicitly.

eg5: `scanf ("%d,%d", &x, &y);` // this will scan two integers separated by a comma

eg6: `scanf ("%d;%d", &x, &y);` // this will scan two integers separated by a semicolon

## (Minimal) Structure of a C Program

① #include <stdio.h>  standard input-output library

② int main(){

/\* series of instructions \*/

}

NOT studio.h

printf, scanf etc. are  
defined in stdio.h  
library

### Points to note:

- Every stmt in C should end with a semicolon
- int main() is the entry point for a C prog
- Input stmts, output stmts, pure expr<sup>ns</sup> and assignment stmts are all called expr<sup>n</sup> stmts, and they always return a value. No other kinds of stmts do.